effort that Weber, Braginsky, and my other friends and colleagues had put into bars, and also because I had become convinced that gravitational radiation has the potential to produce a revolution in our knowledge of the Universe.

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To understand the revolution that the detection and deciphering of gravitational waves might bring, let us recall the details of a previous revolution: the one created by the development of X-ray and radio telescopes (Chapters 8 and 9).

In the 1930s, before the advent of radio astronomy and X-ray astronomy, our knowledge of the Universe came almost entirely from light. Light showed it to be a serene and quiescent Universe, a Universe dominated by stars and planets that wheel smoothly in their orbits, shining steadily and requiring millions or billions of years to change in discernible ways.

This tranquil view of the Universe was shattered, in the 1950s, 1960s, and 1970s, when radio-wave and X-ray observations showed us our Universe's violent side: jets ejected from galactic nuclei, quasars with fluctuating luminosities far brighter than our galaxy, pulsars with intense beams shining off their surfaces and rotating at high speeds. The brightest objects seen by optical telescopes were the Sun, the planets, and a few nearby, quiescent stars. The brightest objects seen by radio telescopes were violent explosions in the cores of distant galaxies, powered (presumably) by gigantic black holes. The brightest objects seen by X-ray telescopes were small black holes and neutron stars accreting hot gas from binary companions.

What was it about radio waves and X-rays that enabled them to create such a spectacular revolution? The key was the fact that they brought us very different kinds of information than is brought by light: Light, with its wavelength of a half micron, was emitted primarily by hot atoms residing in the atmospheres of stars and planets, and it thus taught us about those atmospheres. The radio waves, with their 10million- fold greater wavelengths, were emitted primarily by near-light-speed electrons spiraling in magnetic fields, and they thus taught us about the magnetized jets